

Matthew Chesnut

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Education

University of Maryland | College Park, MD

- B.S. Mechanical Engineering — Expected 2027
- Engineering Honors Program (EHP)

Experience

Project Engineer Intern | **Pillar Construction** | Alexandria, VA

June 2026 – August 2026

- Performed quantity takeoffs, cost estimation, and proposal development using OST & QBID for projects totaling over \$9M. Worked with & designed exterior insulation and finish systems (EIFS), rainscreen facades, sub-framing, plaster air barriers, & cement panels.
- Used 3D laser-scanning hardware (Zoller+Fröhlich) across the \$100M+ Northern Virginia Science Center landmark project to capture existing site conditions; processed raw point-cloud data within BIM/VDC environments to perform spatial clash detection and verify strict structural tolerances against the building's irregular geometric architecture.
- Performed clash detection and model corrections for Microsoft IAD12 & 13 data center buildings (\$3B+).
- Managed daily on-site logistics, cross-functional subcontractor workflows, and design-to-field constraints for the 70,000 sq. ft. Science Center building, tracking project operations from initial estimation through physical field execution to deliver a comprehensive final capstone project.

Stem Tutor | **College of Southern Maryland** | LaPlata, MD | Prince Frederick, MD

January 2025 – August 2025

- Tutored students in math through calculus III, calculus-based physics, and engineering courses.
- Assisted students in understanding concepts and problem-solving strategies. Built strong connections with students to create a supportive learning environment. Adapted teaching methods to meet individual learning needs.

Auction Operations Team Lead | **Bunting Auctions** | Owings, MD

April 2022 – August 2025

- Supervised and trained a team of 4 photographers, ensuring efficiency and quality standards for auction listings.
- Assisted customers with pickups, inquiries, and consignor contracts, driving new sign-ups and maintaining strong client relationships.

Projects

- **3-Axis Robotic Arm:** Designed and fabricated a robotic arm utilizing stepper motors, embedded motion control, and inverse kinematics for repeatable precision positioning. Developed Python-based software with real-time 3D visualization and serial communication for interactive robotic manipulation.
- **Geothermal Heat Pump Cycle:** Designed & modeled a dual mode 12.3kW thermal system capable of high efficiency space conditioning (COPs: 6.805 winter / 4.504 summer) using R-454B refrigerant. Performed thermodynamic cycle analysis and component sizing for both summer and winter operating conditions.
- **GPS Speedometer:** Built an Arduino-based GPS device to display real-time speed, satellites, and coordinates, programmed in C with a custom 3D-printed enclosure.

Awards & Achievements

- Eagle Scout, Engineering Honors Program, Dean's List all semesters

Technical Skills

CAD: SolidWorks, AutoCAD, Onshape, Revit, Navisworks | **Programming:** Python, MATLAB, C/C++ | **Manufacturing:** MIG Welding, Machining, 3D Printing, Fabrication | **Electronics:** Circuit Design & Analysis, Arduino | **Engineering:** Embedded Systems, Data Analysis, Simulation